

Project Description: Car Rental Marketplace App

Learning Outcomes:

- Working with a database
- Working with authentication services
- Creating an application of medium-complexity

Group size:

- You may work in a group of maximum 3 people

Submission:

For your submission to be graded, submit

1. Submit the zip file of your code to the project dropbox by 11:59pm on April 13, 2025
2. In the week of April 14-18, book a time with me to demonstrate your application. If you are working in a group, all members must attend and you should explain how you divided work.

In this project, you will build a clone of the Turo app. Turo is a marketplace for people who are looking to rent cars can connect with people who have cars to rent.

Your project must consist of 2 apps:

1. **Car Owner app:** Used by the “owner” to post the cars they **want** available to rent
2. **Renter app:** Used by customers to search for cars to rent

Problem Description

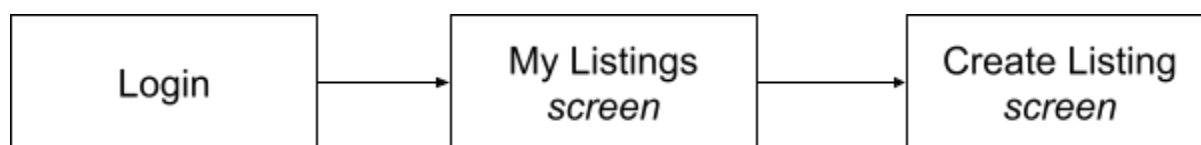
User Account Management

- User accounts must be implemented with [Firebase Authentication](#).
- Both apps must have screens to enable user login (you can reuse the same screen)
- Both apps must provide a way for the user to [logout](#).
- The account system must be able to distinguish between users who are [owners](#) versus [users](#).
- For simplicity, do **NOT** implement user account signup screens. Instead, manually create the user using the Firebase Authentication Console (website)

Data Persistence

- All relevant data must be persisted to [Firebase Firestore](#).
- Both apps should read from the *same* Firestore database.
- Design your collections in a way that makes it easy to share data between both applications.

Owner App Features



1/ User login screen

1. The initial screen of the app is a **login** screen.
2. After logging in, the user can **Create a Listing** or **Manage Bookings**

2/ Create a Listing Screen

Provide a screen that has a form for the user to create a listing for their car.

The form must capture:

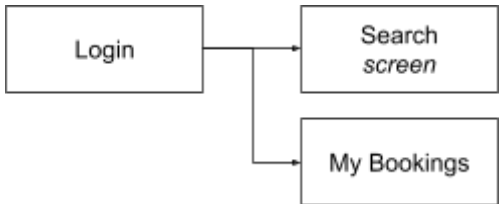
1. Car model (Honda Civic) and license plate.
2. Cost to rent the car
3. A photo/image of the car. [The photo should be a web url.](#)
4. The *city* and *address* of the item/service. An example is:
 - City: Winnipeg
 - Address: 1485 Portage Avenue

3/ My Listings Screen

1. Provide a screen where the owner can manage their bookings.
2. For each booking, show:
 - Information to identify the car is being booked (including price)
 - Name of user who made the booking
 - Booking confirmation code
 - CANCEL BOOKING button. Pressing this button cancels the booking and removes it from the renter's booking list.

Renter App Features

This app enables a user to search for and book a car. For simplicity, a user can only have 1 booking at a time.



Login Screen

1. Provide a login screen. You may reuse your login screen from the owner app.
2. After successful login, navigate the user to either the Search Screen or My Bookings screen. (Use a Tab Navigator)

My Bookings Screen

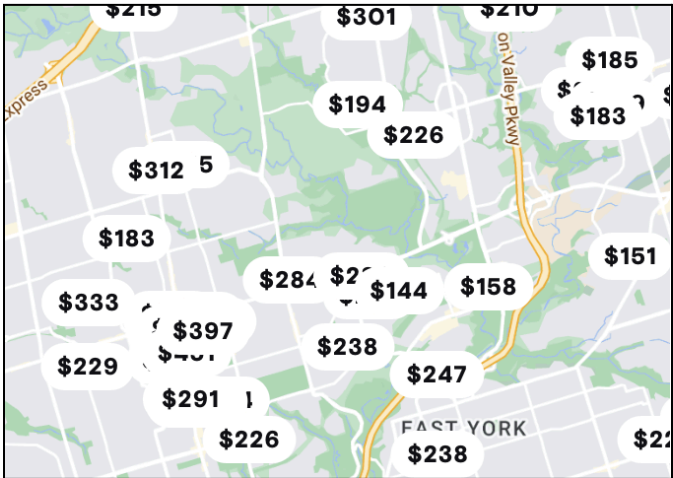
If the user has a booking:

- display the booking’s confirmation code, item information, and pickup location (address and city of the item).
- Provide a “cancel” button. When pressed, make the necessary database updates to cancel the booking.

If there is no booking, then show an appropriate error message.

Search Screen

The screen must display a MapView of all listings in the same city as the user’s location:



1. To retrieve the user’s location, choose from these options:
 - a. OPTION 1: When the screen loads, use the geocoding libraries to automatically retrieve the device location
 - b. OPTION 2: Provide a textbox for the user to enter their city.
2. Each listing must be displayed as a map marker. The map marker must provide information about the listing price. Options for implementation:
 - a. Use the default Marker, but provide the price in the marker’s callout
 - b. Use a custom Marker to display the price
3. When a Map Marker is pressed, the screen should display a summary of the selected listing
The summary must show
 - a. Photo of the car

- b. Car information (model, license plate)
- c. Price
- d. Owner name
- e. BOOK NOW button

Example user interface:



4. Booking an item: For simplicity, the user can only ever have 1 booking at a time. Therefore, when the BOOK NOW button is pressed
- a. Replace any existing bookings with a new booking
 - b. auto generate a booking confirmation code
 - c. display a **success message** and show the **address** and **city** of the item
 - d. update the relevant database collections.

When the user returns to the My Bookings page, the booking should appear.

END OF ASSESSMENT